

# The playful chronicles: investigating adolescents' video game engagement with an entertainment flair<sup>☆</sup>

A. Balamurugan<sup>a,\*</sup>, Sudeep Varshney<sup>b</sup>, K. Raghuveer<sup>c</sup>, M.Kishore Kumar<sup>d</sup>, Siddharth Misra<sup>e</sup>, M.Ramkumar Prabhu<sup>f</sup>

<sup>a</sup> Department of Management Studies, School of Commerce and Management, Bharath Institute of Higher Education and Research, Agarm-then Road, Selayur, Chennai 73 Tamilnadu, India

<sup>b</sup> Department of Computer Science and Engineering, School of Computing Science and Engineering, Sharda University, Greater Noida, Uttar Pradesh, 201310, India

<sup>c</sup> School of Management, Siddhartha Academy of Higher Education-Deemed to be University, Kanuru, Vijayawada-520007, A.P., India

<sup>d</sup> Department of CSE (Data Science), CMR Technical Campus, Hyderabad 501401, Telangana, India

<sup>e</sup> Assistant Professor Grade I, XIM University, Bhubaneswar, India

<sup>f</sup> Department of ECE, Saveetha School of Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, Tamil Nadu, India

## ARTICLE INFO

### Keywords:

Video Game

Students

Video game addiction

Habits

## ABSTRACT

In the age of rapidly growing digital entertainment through video games, understanding their impact on students and identifying related factors is crucial for educational and developmental considerations. Video game use has an influence on social conduct, academic achievement, and general well-being in addition to being merely amusing. The study's aim is to provide insights into the intricate nature of video game engagement among the demographic by examining variables like social media usage, parental involvement, and device ownership. To determine the extent to which students use video games and factors influence their gaming habits. The study utilized a Video Game Addiction Scale alongside a comprehensive survey. Participants were extended to 200 students enrolled in 8th and 9th grades across multiple schools to participate in the survey. To explore relationships and associations among variables, statistical analyses are conducted, such as inferential tests and descriptive statistics. The findings provide insight into potential variables that might influence both positive and negative features of video game participation. Participants can develop an environment that motivates students to play video games responsibly and in a balanced manner by acknowledging the complexity of the elements that influence gaming behaviour.

## 1. Introduction

Long-term video game play has several negative implications in addition to its advantages, which include competency, enjoyment and fun. It has discovered a number of psychological influence on behavior and health that are similar to those of excessive watching television. The integration of multiple methodologies within the gaming period facilitates researchers in conducting a comprehensive analysis of the intricate dynamics between video game participation and student behavior. These strategies can be utilized by educators and policymakers to create customized interventions aimed at fostering responsible gaming habits and enhance the educational advantages of video games for pupils.

Video game technology has advanced to the point where it can create entirely new worlds for players to explore [1]. Players can become addicted to them and engage in risky behaviors in the virtual world that they would never dream of committing in real life. To prevent their children from being addicted to video games, parents should monitor their screen time and set limits [2]. Additionally, it is the responsibility of parents and teenagers to determine whether the game's content is suitable. The violent media material influences users to behave violently, however, the main distinction between violent media and violent games is the user's interaction with the virtual environment of the former [3]. People seem to be worried anytime a new technological development emerges. As the case with radio, comic books, and

<sup>☆</sup> This article is part of a special issue entitled: 'ENT for Tomorrow' published in Entertainment Computing.

\* Corresponding author.

E-mail addresses: [drabala72@gmail.com](mailto:drabala72@gmail.com) (A. Balamurugan), [Sudeep149@gmail.com](mailto:Sudeep149@gmail.com) (S. Varshney), [r.katragadda@vrsiddhartha.ac.in](mailto:r.katragadda@vrsiddhartha.ac.in) (K. Raghuveer), [Kishore.mamidala@gmail.com](mailto:Kishore.mamidala@gmail.com) (M.Kishore Kumar), [siddharth.misra@bgu.ac.in](mailto:siddharth.misra@bgu.ac.in) (S. Misra), [ramkumarprabhu@gmail.com](mailto:ramkumarprabhu@gmail.com) (M.Ramkumar Prabhu).

<https://doi.org/10.1016/j.entcom.2025.101032>

Received 20 January 2024; Received in revised form 29 June 2024; Accepted 29 September 2025

Available online 10 October 2025

1875-9521/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

newspapers, negative connotations are attached to them. There was a theory that the public exposure of complete criminal cases in newspapers encouraged offenders to commit and then evade punishment [4]. Comic books were the first and most important evidence in the investigation of every tragedy that occurred in someone's life, and radio was accused of being a source of catharsis for individuals listening to plays with criminal tales [5]. Fig. 1. represents the video game elements.

People respond to certain video games, such as online multiplayer games, because they provide the social interaction features. The consideration of time spent is an issue when communicating with individuals from all over the world [6]. This is based on the findings of several sociological and scientific studies. Video games expose players to violent material than other violent media because, after dying in-game, they can restart the game several times to reach certain objectives [7]. There is a lengthy history of political contestation and social instability stoked by violent video games. The persuasive and educational power of video games has led to their widespread adoption by military enterprises throughout the globe for training and recruitment. However, the problem starts, when these games are distributed to the wider public as well [8]. At the same time it was promoted as an amusement game, Full Spectrum Warrior was used for military training. Due to advancements in video game creation and more realistic visuals for presentation, the video game business is gaining popularity for both military training and the entertainment sector at the same time. Politics of terror and violence are also used in these operations. video games are stepping as a mediator in this digital space [9]. The complex nature of conflict necessitates mental and financial resources. Recognizing the significance of understanding the viewpoints and interactions of teenagers with the virtual realm, to link the space between academic discourse and the active world of video game culture in the Playful Chronicles, explore how the fields of entertainment and study [10] come together to explain the complex web of teenage video game participation. The aim of the article to provide insights into the complex nature of video game engagement among the demographic by examining variables like social media usage, parental involvement and device ownership. Examining video game participation among eighth- and ninth-grade children, with a focus on social media use, parental involvement, and device ownership. The goal is to provide information for encouraging responsible gaming practices.

## 2. Related works

The study [11] suggested rules are not making an impact because

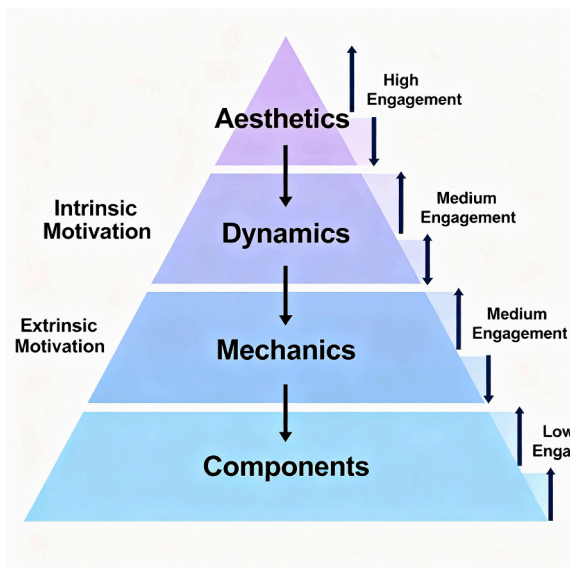


Fig. 1. Video game Elements.

lawmakers are caught between the competing goals of helping the video game business thrive and minimizing the prevalence of video game addiction among gamers. An all-encompassing strategy to reduce gaming addiction does not exist, and the problem was largely to blame. The study [12] suggested adult gamers see monetization as an integral aspect of the game experience. There seems to be a need for greater openness with the monetization processes, however certain approaches do create ethical concerns even among gamers. The study [13] explained their addictive nature and created the optimal conditions for maximum appeal and excitement, the research examined the changes in physical quantities that occur during the gameplay of arcade games. Research has shown that arcade games' fundamental allure was the jerk dynamics that are relevant to the gaming process. The study [14] examined the steps involved in building a data pipeline that can identify the key characteristics of intelligent agents that can be trained in game-based settings. The practice to create intelligent agents that can stand for the actions and thought processes of whole groups or even individuals. The article [15] provided a brief overview of current events and explores their potential future effects on the fashion and gaming industries. The main idea was young designers can utilize environment customization, and offices to break into the creative sectors. The study [16] summarized and assessed which is known about gamification and serious games in the workplace by reviewing relevant literature. Many businesses are hesitant to embrace gamification because of issues like demography, ethics and a track record of unsuccessful efforts at the strategy, in addition to the difficulties of implementing change inside organizational hierarchies. The study [17] examined football video game aesthetics via the lens of the mechanics-dynamics-aesthetics (MDA) paradigm. To employ qualitative theme analysis and semi-structured interviews to gather and analyzed the data. The study [18] examined the technological advancements in computing power and the proliferation of high-speed internet have contributed to the exponential growth of the gaming industry around the world. People who play video games spend a lot of time, effort, and resources on them, which can lead to addiction. The study [19] analyzed three video games that serve as case studies for how interactive, non-narrative design elements can be used to depict mental illness in video games. The study argues that video games can do a better job of representing mental illness by considering the games' audiovisual styles, control methods, game objectives, and procedural. The article [20] suggested the Blithe Child's resemblance to classic toys serves many purposes: to illustrate and clarify non-violent gaming mechanics; to mold sentimental bonds between players and their avatars; to provide safe, inviting play environments; and to promote cooperative, imaginative, and expressive playtime habits. The study [21] considered the evolution of community distraction venues such as arcades, coin-operated amusement systems, video games as well as pool halls and the socioeconomic environment of the time linked to the rise of bachelor subculture investigate the historical context surrounding the formation known as arcade culture. The study [22] detailed the findings of two experiments that sought to examine the impact of game modifications on player experience in player immersion. To do this, two games were created with two distinct types of adjustments to modify the game's complexity according to the player's progress. The study [23] suggested a qualitative analysis method was used to research, examine, and compare competing game meanings put out by well-known literary critics. Traditional sports were described and characterized for the research professionalism, gambling, and sports, as well as data from case studies and interviews with professional and amateur sports players, enthusiasts, staff, and a color commentator on the subject of gamer motivations. The article [24] looked at The Guardians: Unite the Realms, a brand new, available smartphone game that promotes healthy habits in real life via the use of in-game prizes. The study [25] explored many ways in which audiovisual elements, control schemes, game objectives, and procedural algorithms depict mental illness in video games. The study [26] aimed to combat the widespread dismissal of video games as lowbrow forms of amusement among scholars and critics. It argues that

video games can tell compelling stories through a unique perspective that differs from films and novels. The study [27] looked at the critical reception of the well-received Star Wars games, which sheds light on a lot of possible considerations, such as the games' audio-visual quality, the accompanying plot, and parts of the gameplay. Additionally, the article delves into the conflicting ideals of quality in video games and Star Wars adaptations. The study [28] evaluated the effects of roles on participants' views of their engagement, competence, and autonomy in remote, mixed-visual-ability research. Participation in the games was enjoyable and difficult regardless of one's visual capabilities and found that intentionally uneven experiences can give rise to egalitarian ones when shared. The study [29] examined the relationship between intractable conflict, empathy, and prejudice, and also discusses how games might inspire empathy. In the latter section of the thesis, the results of the study are summarized and suggestions are made for the gaming industry might make games with a greater effect. The study [30] examined the potential to inform industry strategies for making online games more educational and to help researchers in the fields of behavior and physiology interpret the phenomenon from a variety of angles.

Existing research provides further context on video game engagement, including its effects on addiction, monetization, portrayal of mental illness, historical background and potential educational advantages. Comparisons with these studies have the potential to enhance the understanding of the findings by offering valuable insights into the wider scope of research on video games, encompassing diverse methodology, theoretical frameworks, and conclusions.

### 3. Methodology

Participants in the poll included 200 children who could read and write in English and were enrolled in the eighth and ninth grades. The age range of the responders is 13–16 years old (Table 1). Approximately (57 %) of teenagers are 13 years old, (38.5 %) of kids are 14 years old, (4 %) are fifteen years old, and (0.50 %) are 16 years old. Approximately one-fifth, (21 %) of the participants were single, and (79 %) of them included older or younger siblings in the household. The participant's birth order reveals that the majority (47 %) were firstborn, (11 %) were middleborn, and (29 %) were lastborn. Table 2.

#### 3.1. Structured survey

Each variable had research information gathered for the item. The sociodemographic traits, the frequency, skill set, and perceptions of video gaming, the parents' perspectives, the utilization of social networking by students and young parents, and video gaming.

**Table 1**  
Students' sociodemographic profile.

| Description           | Value (%) |
|-----------------------|-----------|
| <b>Age group</b>      |           |
| 13                    | 114(57)   |
| 14                    | 77(38.5)  |
| 15                    | 8(4)      |
| 16                    | 1(0.5)    |
| <b>Gender</b>         |           |
| Male                  | 123(61.5) |
| Female                | 77(38.5)  |
| <b>Type of family</b> |           |
| Joint                 | 24(12)    |
| Nuclear               | 172(86)   |
| Single-member         | 4(2)      |
| <b>Income</b>         |           |
| <10000                | 104(52)   |
| Above 10,000          | 96(48)    |

**Table 2**  
Relationships between personal and family characteristics.

| Variables                     |              | N (%)          | Geometric Standard Deviation | Geometric mean score | p-value |
|-------------------------------|--------------|----------------|------------------------------|----------------------|---------|
| Twitter                       | Yes          | 34<br>(22.67)  | 55.4 <sup>a</sup>            | 1.14                 | 0.002   |
|                               | No           | 116<br>(77.33) | 44.3 <sup>b</sup>            | 1.6                  |         |
| Parents video gaming          | At least one | 70<br>(46.67)  | 46.5                         | 1.85                 | 0.415   |
|                               | Neither      | 80<br>(53.33)  | 44.8                         | 1.21                 |         |
| Social media usage of parents | At least one | 129<br>(86)    | 44.8                         | 1.28                 | 0.473   |
|                               | Neither      | 21 (14)        | 43.3                         | 1.12                 |         |
| Social media usage of sibling | Yes          | 92<br>(61.33)  | 45.9                         | 1.5                  | 0.212   |
|                               | No           | 58<br>(38.67)  | 44.8                         | 1.1                  |         |
| Television ownership          | Yes          | 9 (6)          | 46.7                         | 1.53                 | 0.724   |
|                               | No           | 141<br>(94)    | 45.3                         | 1.12                 |         |
| Room ownership                | Yes          | 134<br>(89.33) | 42.4                         | 1.8                  | 0.527   |
|                               | No           | 16<br>(10.67)  | 46.9                         | 1.6                  |         |
| Smartphone ownership          | Yes          | 95<br>(63.33)  | 46.2                         | 1.43                 | 0.034   |
|                               | No           | 55<br>(36.67)  | 44.1                         | 1.78                 |         |
| Sibling video gaming          | Yes          | 88<br>(58.67)  | 45.8                         | 1.1                  | 0.647   |
|                               | No           | 62<br>(41.33)  | 44.6                         | 1.2                  |         |
| Facebook                      | Yes          | 82<br>(54.67)  | 48.7 <sup>a</sup>            | 1.52                 | 0.002   |
|                               | No           | 68<br>(45.33)  | 42.8 <sup>b</sup>            | 1.28                 |         |
| Social media usage of student | Yes          | 103<br>(68.7)  | 47.2                         | 1.5                  | 0.035   |
|                               | No           | 47<br>(31.3)   | 43.3                         | 1.36                 |         |
| Instagram                     | Yes          | 99 (66)        | 47.4                         | 1.23                 | 0.072   |
|                               | No           | 51 (34)        | 43.8                         | 1.57                 |         |
| Tablet ownership              | Yes          | 125<br>(83.33) | 45.8                         | 1.96                 | 0.267   |
|                               | No           | 25<br>(16.67)  | 45.1                         | 1.17                 |         |
| Computer ownership            | Yes          | 123<br>(82)    | 46.3                         | 1.52                 | 0.723   |
|                               | No           | 27 (18)        | 44.8                         | 1.48                 |         |
| Game console ownership        | Yes          | 12 (8)         | 51.2 <sup>a</sup>            | 1.54                 | —       |
|                               | No           | 138<br>(92)    | 44.8 <sup>b</sup>            | 1.86                 |         |

#### 3.2. Study tools

To evaluate the length of time spent playing video games, how often they are used, what people do for fun, and their sentiments towards video games were assessed, with all responses measured using a four-point rating system. Response A – receives a value of 2, reaction B – receives a value of 5, response C – receives a value of 10, and response D – receives a value of 0. Psychiatry social work, clinical psychology, nursing, and psychiatry for children and adolescent faculty members performed face validation to build cultural sensitivity. The scale's results suggest that students are nonusers from 0 to 20, that users have control from 25 to 40, that students are excessively using video games from 50 to 70, and that they are dependent from 80 to 110. The following represents a checklist that students employ to evaluate their approach to learning. The checklist includes 30 objects, every after two replies, yes and no. "Yes" responses were recorded as 1 in the current study, whereas "no" responses were coded as 0. Seven domains were identified from the

comparable topics on the checklist. These consist of two items for time management, three for taking notes, four for reading and homework assignments, two for memory, five for group study, two for classroom activities, and two for leisure time activities. A lower score denotes greater issues with study habits. The social and demographic characteristics of the students, their entertainment habits, and the incidence of potential gaming addiction as determined by the video game dependency scale were described using a descriptive approach.

### 3.3. Children's video game addiction scale

Students who played video games were subjected to VASC. This scale consists of 21 elements. Every category on the Likert-type scoring scale has the following markings: none = 1, seldom = 2, occasionally = 3, usually = 4, or always = 5. The ratings to obtain every component were added up to determine the overall score for every student. The total rating ranges between 20 and 110. The acceptable limit score for addiction was 100, as shown by the literature. Fig. 2 shows the addition scales for video games. The term Score on video game addiction scale of the study participants describes how participants' vulnerability to video game addictions is measured quantitatively in a study. Generally measured with standardized instruments such as the Internet Gaming Disorder Scale (IGDS) or the video Game Addiction Test (VAT), participants answer questions on the degree to which they engage in game-related behaviours and behaviours. According to Fig. 2, 20 % of students play video games excessively, 18 % of students use them for light enjoyment, and 44.5 % of students do not use them at all. Addiction (video game addiction) affects about 17.5 % of the student body.

### 3.4. Statistical analysis

The following statistical methods have been generated for the entire example: The categorical variables were frequency ranges and the group comparisons were made using the proportional probabilities of the

categorical data using the inferential tests and descriptive statistics. The geometric average and geometrical standard deviations were utilized as the VASC total scores were log-normally distributed. The log-transformed VASC overall score values have been compared using each group's independent *t*-test using an ANOVA. Using Cronbach's alpha analysis, the VASC's dependability was evaluated; the result was 0.90. The application of software known as the Statistical Software Programmed for the Social Sciences (SPSS) 21.0 is utilized to perform statistical analysis. P-values less than 0.05 were regarded as significant in statistical terms. The statistical techniques employed encompass proportional probabilities for categorical variables, descriptive statistics for data summarization, geometric mean and geometric standard deviation for log-normally distributed data, independent *t*-tests and ANOVA for group comparisons.

## 4. Results

In the fifth and sixth grades at these schools, there were a total of 200 pupils. Every parent who consented to take part finished the survey in its entirety. Due to their ongoing medical conditions, 14 children were not allowed to participate. As a result of the 180 students in excellent physical condition that were part of the study, 150(75 %) said they were games. Each student who played video games finished the VASC. Addicts of video games were the four children whose scores were higher than 90. It involved a group of male students. Throughout the entire collection, the VGA speed is estimated with a value of 1.6 %.

In a gender-specific analysis, the VGA ratio for male students came out to be 3.1 %. In addition to the girls, there was not a single gamer. Within the video gamer sample as a whole, the median VASC- Geometric Mean Score (VASC-GMS) was  $(45.3 \pm 1.4)$ . Table 1 depicts the association between socio-demographic variables and VASC-GMS. It was greater in men  $(50.5 \pm 1.3)$  than women  $(40.4 \pm 1.4)$  and  $(p < 0.001)$  as well as significantly greater in pupils whose BMI is more than 95 % percentile  $(p = 0.022)$ . Table 3 shows the relationships between video

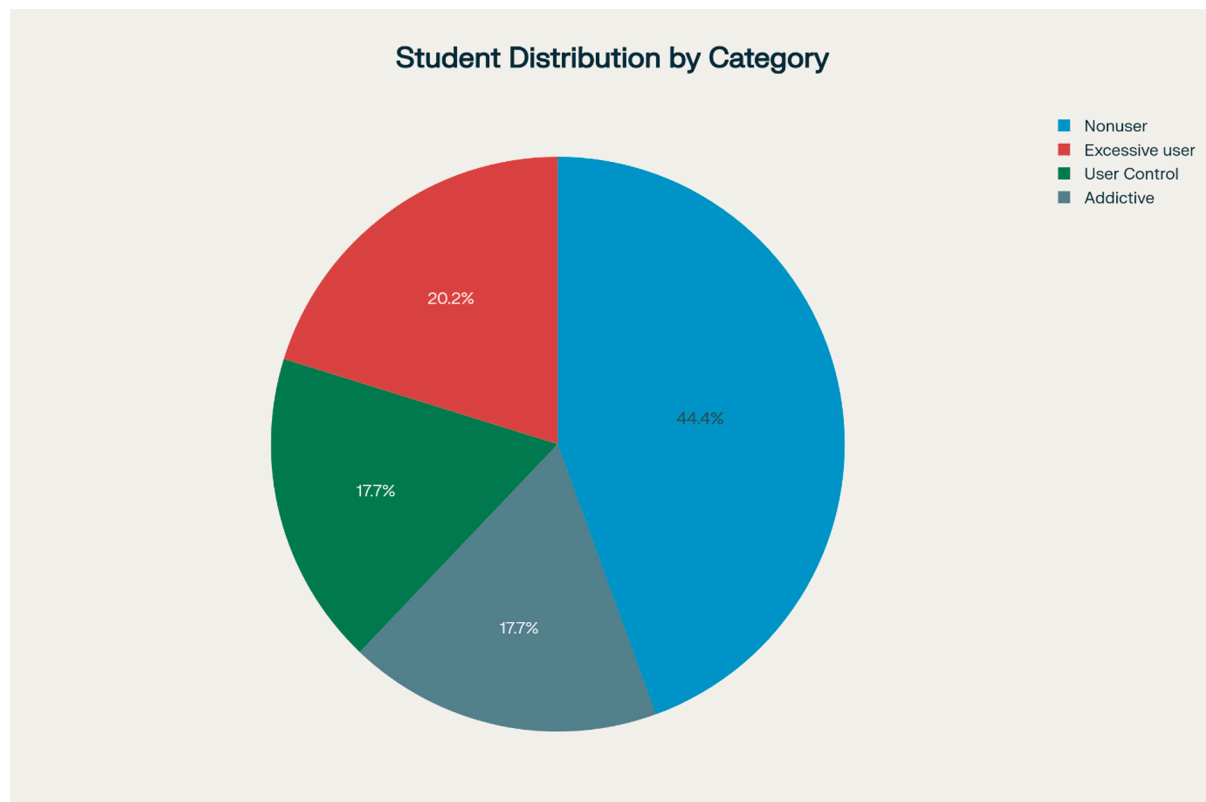


Fig. 2. Scores on the video game addiction scale.



**Table 3**  
Associations between video game habits and VASC-GMS.

| Variables               | Total number (%) | GM Value          | Geometric standard deviation | p-value |
|-------------------------|------------------|-------------------|------------------------------|---------|
| <b>Style of gaming</b>  |                  |                   |                              |         |
| Offline                 | Yes 21(14)       | 47.2              | 1.7                          | 0.002   |
|                         | No 129 (86)      | 45.1              | 1.6                          |         |
| Web-based               | Yes 49 (32.67)   | 41.5 <sup>a</sup> | 1.5                          | 0.315   |
|                         | No 101 (67.33)   | 47.8 <sup>b</sup> | 1.4                          |         |
| Online                  | Yes 58(38.67)    | 49.8 <sup>a</sup> | 1.8                          | <0.001  |
|                         | No 92 (61.33)    | 43.1 <sup>b</sup> | 1.3                          |         |
| <b>Gaming platforms</b> |                  |                   |                              |         |
| Game console            | Yes 15 (10)      | 51.2 <sup>a</sup> | 1.3                          | 0.028   |
|                         | No 135 (90)      | 44.6 <sup>b</sup> | 1.1                          |         |
| Mobile phone            | Yes 116 (77.33)  | 45.2              | 1.2                          | 0.069   |
|                         | No 34 (22.67)    | 41.63             | 1.8                          |         |
| Computer                | Yes 88 (58.67)   | 46.25             | 1.5                          | 0.430   |
|                         | No 62(41.33)     | 44.48             | 1.23                         |         |
| Tablet                  | Yes 77 (51.33)   | 45.9              | 1.5                          | 0.970   |
|                         | No 73 (48.67)    | 45.86             | 1.2                          |         |
| <b>Number of Player</b> |                  |                   |                              |         |
| Multiplayer             | Yes 67 (44.67)   | 48.5 <sup>a</sup> | 1.9                          | 0.016   |
|                         | No 83 (55.33)    | 43.6              | 1.8                          |         |
| Single                  | Yes 17 (11.33)   | 44.6              | 1.3                          | 0.413   |
|                         | No 133 (88.67)   | 46.2              | 1.2                          |         |
| <b>Genres of game</b>   |                  |                   |                              |         |
| Casual                  | Yes 36 (24)      | 40.7 <sup>a</sup> | 1.2                          | 0.005   |
|                         | No 114 (76)      | 46.7              | 1.2                          |         |
| Mind                    | Yes 48(32)       | 39.4 <sup>a</sup> | 1.6                          | 0.006   |
|                         | No 102(68)       | 46.1 <sup>b</sup> | 1.5                          |         |
| Race                    | Yes 41 (27.33)   | 47.7              | 1.8                          | 0.346   |
|                         | No 109 (72.67)   | 47.6              | 1.9                          |         |
| Sport                   | Yes 21(14)       | 49.2              | 1.2                          | 0.155   |
|                         | No 129 (86)      | 44.8              | 1.4                          |         |
| Action-Adventure        | Yes 58 (38.67)   | 48.8              | 1.7                          | 0.051   |
|                         | No 92 (61.33)    | 44.7              | 1.4                          |         |
| Strategy                | Yes 56 (37.33)   | 48.4              | 1.1                          | 0.034   |
|                         | No 94 (62.67)    | 44.1 <sup>b</sup> | 1.5                          |         |
| War                     | Yes 34 (22.67)   | 52.2 <sup>a</sup> | 1.2                          | 0<.001  |
|                         | No 116 (77.33)   | 43.3              | 1.3                          |         |

(Note: GM Value – Geometric mean value).

game patterns and VASC-GMS. Pupils who used gaming consoles and played online games performed better than their classmates ( $p = 0.028$  and  $0.002$ , respectively). Pupils who engaged in web-based gaming exhibited considerably ( $< 0.001$ ) reduced scores. VASC-GMS discovered that students who played multiplayer games had higher ( $p = 0.016$ ) when taking into the number of participants. In the comparison of different gaming genres, the VASC-GMS for strategy and war games was higher ( $p = 0.034$  and  $p < 0.001$ ). It lowers in students who played mind games and casual games ( $p = 0.006$  and  $p = 0.004$ ).

To distinguish between situations when the same % yields distinct Geometric Mean Scores, the subscripts “a” and “b” are employed. The associations between video gaming activities and VASC-GMS are shown in Table 4. Playing video games (VASC-GMS) for more than four hours on both days of the week was linked to higher scores in students ( $p < .001$  and  $p < .001$ ). Students who put off their daily assignments to play video games did not do as well on assessments as those who did not ( $p < .001$ ). A student’s addiction score grows compared to their classmates if they frequently neglect their homework ( $p = 0.001$ ). It was conducted into a student’s reaction to the restrictions placed by parents on video games. The VASC-GMS a student continues to maintain a high ( $p = .002$ ).

#### 4.1. Most popular Type of video game

Fig 3 displays participation demographics for the majority of popular video games at the elementary and high school levels. The majority of secondary school participants preferred battle and adventure games ( $f = 21$ ) and sports games like football and automobile racing ( $f = 24$ ). The most popular genres among high school participants were battle and

**Table 4**  
Relationships between VASC-GMS and video game behaviours.

| Variables                         | Total number (%)     | GM Value          | Geometric standard deviation | p-value |
|-----------------------------------|----------------------|-------------------|------------------------------|---------|
| Purchasing a game                 | No 125 (83.33)       | 44.2 <sup>b</sup> | 1.5                          | 0.646   |
|                                   | Yes 25 (16.67)       | 52.7 <sup>a</sup> | 1.5                          |         |
| Engaging in games at home         | No 20 (13.33)        | 43.3              | 1.2                          | 0.202   |
|                                   | Yes 130 (86.67)      | 45.5              | 1.6                          |         |
| VG time on weekends               | 2–4 h 45 (30)        | 45.8 <sup>a</sup> | 1.4                          | 0<.001  |
|                                   | <2 h 76 (50.67)      | 42.2 <sup>a</sup> | 1.5                          |         |
|                                   | >4 h 29 (19.33)      | 53.1 <sup>b</sup> | 1.5                          | 0.047   |
| Playing at gamer saloon           | Yes 5 (3.33)         | 54.4              | 1.8                          |         |
|                                   | No 145 (96.67)       | 54.2              | 1.7                          | 0<.001  |
| VG time on weekdays               | <2 h 119 (79.33)     | 43.8 <sup>a</sup> | 1.7                          |         |
|                                   | 2–4 h 22 (14.67)     | 51.2 <sup>a</sup> | 1.8                          | 0.332   |
|                                   | >4 h 9 (6)           | 55.8 <sup>b</sup> | 1.6                          |         |
| Playing at friend’s home          | Yes 12 (8)           | 43.1              | 1.2                          | —       |
|                                   | No 138 (92)          | 45.2              | 1.9                          |         |
| Reaction of students              | Obey 102 (68)        | 43.3 <sup>a</sup> | 1.5                          | 0.001   |
|                                   | Obstinate 48(32)     | 49.2 <sup>b</sup> | 1.2                          |         |
| Skip homework to play             | Always 11 (7.33)     | 54.6 <sup>a</sup> | 1.9                          | 0.002   |
|                                   | Sometimes 25 (16.67) | 48.6 <sup>b</sup> | 1.1                          |         |
|                                   | Rarely 48(32)        | 48.7 <sup>b</sup> | 1.2                          | 0.125   |
|                                   | Never 66 (44)        | 42.4 <sup>b</sup> | 1.6                          |         |
| Delay of daily needs              | Always 5 (3.33)      | 57.1 <sup>a</sup> | 1.1                          | 0.002   |
|                                   | Sometimes 25 (16.67) | 49.1 <sup>b</sup> | 1.1                          |         |
|                                   | Rarely 48 (32)       | 47.1 <sup>b</sup> | 1.3                          | 0.001   |
|                                   | Never 72 (48)        | 41.8 <sup>b</sup> | 1.3                          |         |
| Prefer VG to activities they like | Always 9 (6)         | 45.8              | 1.3                          | 0.125   |
|                                   | Sometimes 12(8)      | 48.3              | 1.3                          |         |
|                                   | Rarely 66 (44)       | 45.5              | 1.1                          | 0.002   |
|                                   | Never 63 (42)        | 42.1              | 1.1                          |         |

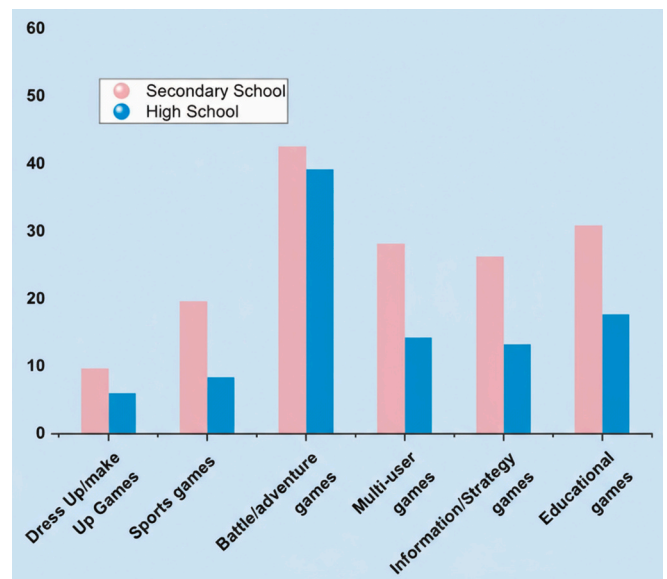


Fig. 3. Type of video game most played.

adventure games ( $f = 55$ ), sports games like football, and automobile racing ( $f = 40$ ), and information and strategic games ( $f = 40$ ).

#### 4.2. Engaged in Recreational digital activities

The following percentage is the share of students in school, usually in the 8th through 9th years, who utilized a computer for non-academic purposes, such as playing video games or other computer games, or for purposes other than study. This indicator measures how common it is for students in school to play video or computer games and use computers in general for non-academic purposes. It offers information about how students use technology for leisure and recreation. Tracking this proportion makes easier for educators and legislators to comprehend how digital media affects students' everyday lives and allows for more focused interventions or curriculum modifications that support a balanced approach to technology use as shown in Fig. 4.

#### 4.3. Discussion

The results of research involving fifth- and sixth-grade kids are presented in the section that follows, providing insight into the connection between video gaming behaviours, physical health, and academic success. Three hundred and eleven parents gave their permission to participate in the study, and each parent filled out the survey. Interestingly, 14 children who had persistent medical issues were not included. Of the 180 pupils who were in good physical health, 82.5 % said they played video games, and four kids showed signs of addiction to gaming. When the study dives into gender-specific studies, it finds that male students have a video game addiction ratio of 3.1 % while female gamers have none. Significant differences in VASC-GMS are noted between male and female students, as well as those whose BMI is higher than the 95th percentile. Academic performance is also influenced by the kind of video games that are played; strategy and war games are connected to better VASC-GMS scores, whereas mind and casual games are linked to lower scores. The links between patterns, behaviours, and VASC-GMS in video games are further explained in Tables 3 and 4. Interestingly, longer gaming sessions and disregard for everyday obligations are linked to greater addiction ratings. In the last part, the most popular video game genres among elementary and high school students are shown, with a focus on the popularity of sports, strategy, and war and adventure games. The study offers an examination of the relationship between fifth- and sixth-grade students' video game habits, academic achievement, and sociodemographic characteristics. It also offers insightful information to educators and legislators who are trying to comprehend and address the effects of digital media on the education of pupils.

#### 5. Conclusion

To investigate factors such as social media usage, parental participation, and device ownership to disentangle the complex dynamics of video game engagement among children in the eighth and ninth grades. Using a video Game Addiction Scale and an extensive survey, the results showed noteworthy correlations with several variables and shed light on the varied terrain of video game usage. The findings highlight the necessity of customized approaches to encourage ethical gaming. Educators and parents should use these insights to acknowledge the intricate interaction of factors on kids' video game engagement and establish an atmosphere that supports balanced gaming habits. The study has limitations, especially in its concentration on a single grade range, even if it makes significant contributions. The findings' validity and generalizability may be compromised by factors such as data quality concerns, restricted availability of specific data sources, possible biases within the sample population, and an excessive emphasis on a single grade range. To gain a complex knowledge of the relationship between video game participation and students' well-being, future research should examine a

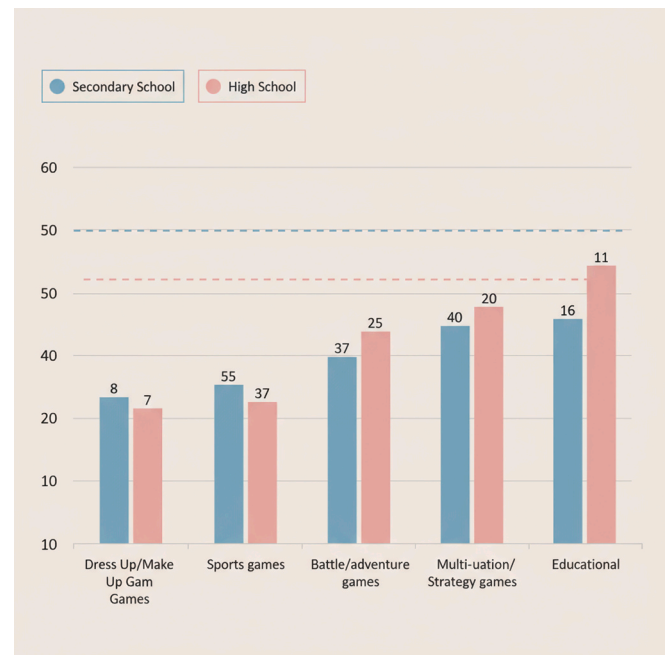


Fig. 4. Video or computer games utilize a computer.

wider range of demographics, go deeper into longitudinal impacts, and take into account other factors. The paper exhibits a structured format, commencing with an introductory section that delineates the study's significance. Subsequently, a Literature Review, Methodology, Results, Discussion, Conclusion, Future Research Directions, and References are presented. Each element of the study contributes to the overall comprehension of video game engagement among students in the eighth and ninth grades. It covers different facets of the subject matter and offers valuable insights that might inform future research endeavours.

#### CRediT authorship contribution statement

**A. Balamurugan:** Writing – review & editing. **Sudeep Varshney:** Writing – review & editing, Funding acquisition, Formal analysis. **K. Raghuveer:** Writing – original draft, Resources, Methodology. **M. Kishore Kumar:** . **Siddharth Misra:** Writing – original draft, Resources, Methodology. **M.Ramkumar Prabhu:** Writing – review & editing, Resources, Methodology, Investigation.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Data availability

The data that has been used is confidential.

#### References

- [1] J.M. Vervoort, New frontiers in futures games: leveraging game sector developments, *Futures* 105 (2019) 174–186, <https://doi.org/10.1016/j.futures.2018.10.005>.
- [2] S. Ponnusamy, V.K.S. Maddala, K. Jayarajan, M. Braveen, R. Walia, P. Krishna, K. Kaliyaperumal, R. Khan, Multi-sensor data and cross-validation technique for merging temporal images for the agricultural performance monitoring system, *J. Food Qual.* 2022 (2022) 9575423.
- [3] N.D. Bowman, S.J. Ahn, L.M. Mercer Kollar, The Paradox of Interactive Media: the Potential for Games and Virtual Reality as Tools for Violence Prevention, *Frontiers in Communication* 5 104 (2020), <https://doi.org/10.3389/fcomm.2020.580965>.

- [4] A. Muscio, The ambiguous role of science and technology in Marvel superhero comics: from their 'Golden Age' to the present-day, *Technol. Forecast. Soc. Chang.* 186 (2023) 122149, <https://doi.org/10.1016/j.techfore.2022.122149>.
- [5] H.U. Warrich, S. Jamil, F.R. Khan, Behavioral Escalation: Games as a Tool of Hybrid War, *Global Mass Communication Review* (2020) 68–82.
- [6] D.A. Cole, E.A. Nick, K.A. Pulliam, Are massively multiplayer online role-playing games healthy or not and why? Preliminary support for a compensatory social interaction model, *Comput. Hum. Behav.* 1 (102) (2020) 57–66, <https://doi.org/10.1016/j.chb.2019.08.012>.
- [7] E. Fontaine, A Comparison of the Effects on Physiological Arousal While playing Violent and Non-Violent games (Doctoral dissertation, The Chicago School of Professional Psychology), 2021.
- [8] R. Godfrey, The politics of consuming war: games, the military-entertainment complex and the spectacle of violence, *J. Mark. Manag.* 38 (7–8) (2022) 661–682, <https://doi.org/10.1080/0267257X.2021.1995025>.
- [9] A. Jungherr, D.B. Schlarb, The extended reach of game engine companies: how companies like epic games and Unity technologies provide platforms for extended reality applications and the metaverse, *Social Media+ Society* 8(2): 20563051221107641 (2022), <https://doi.org/10.1177/20563051221107641>.
- [10] R. Patton, Socially Engaged Art Practices, Tactical Media, and Games: Methods for Civic Participation in Digital Art Education, *InTeaching Civic Participation with Digital Media in Art Education*, Routledge, 2024, pp. 40–55.
- [11] S. Ponnusamy, R.K. Vemuri, P.C.S. Reddy, Puneeth Kumar, J. Ravi, S. Sharma, Deep learning based remote sensing technique for environmental parameter retrieval and data fusion from physical models, *Arab. J. Geosci.* 14 (2021) 1230, <https://doi.org/10.1007/s12517-021-07577-3>.
- [12] T. Tulinén, Entertainment or exploitation: discourses on in-game monetization and gacha mechanics in video games (Master's thesis, T. Tulinén) (2022).
- [13] Y. Gao, C. Liu, N. Gao, M.N. Khalid, H. Iida, Nature of arcade games, *Entertainment Computing* 1;41:100469, (2022), DOI: 10.1016/j.entcom.2021.100469.
- [14] R. Sottillare, C. Ballinger, C. McGroarty, Considerations in the Design of Realistic Agents for Serious Games, In the International FLAIRS Conference Proceedings 36 (2023), <https://doi.org/10.32473/flairs.36.133333>.
- [15] E. Reay, V. Wanick, Skins in the Game: Fashion Branding and Commercial Video Games, In *Reinventing Fashion Retailing: Digitalizing, Gamifying, Entrepreneurship* (2023) (pp. 73–90), Cham: Springer International Publishing, DOI: 10.1007/978-3-031-11185-3\_5.
- [16] K. Larson, Serious games and gamification in the corporate training environment: a literature review, *TechTrends* (2020) 19–28, <https://doi.org/10.1007/s11528-019-00446-7>.
- [17] L. Kim, T.C. Tan, A. Bairner, A Beautiful Game: Interpreting Football game Experiences, *Leis. Sci.* 20 (2023) 19, <https://doi.org/10.1080/01490400.2023.2170497>.
- [18] S.S. Kottalgi, A research proposal: Policy issues of video game addiction (Doctoral dissertation, Doctoral dissertation, Johnson & Wales University), [https://scholarsarchive.jwu.edu/mba\\_student](https://scholarsarchive.jwu.edu/mba_student) (2019).
- [19] S.L. Anderson, Portraying mental illness in games: exploratory case studies for improving interactive depictions, *Loading* 13 (21) (2020) 2033, <https://doi.org/10.7202/1071449ar>.
- [20] E. Reay, Cute, cuddly and completely crushable: Plushies as avatars in video games, *Journal of Gaming & Virtual Worlds* 1;13(2) (2021) 3149, DOI: 10.1386/jgvw\_00033\_1.
- [21] M.R. Skolnik, S. Conway, Tusslers, beatdowns, and brothers: a sociohistorical overview of game arcades and the street fighter community, *Games and Culture* 14 (7–8) (2019) 742–762, <https://doi.org/10.1177/1555412017727687>.
- [22] A. Denisova, P. Cairns, Player experience and deceptive expectations of difficulty adaptation in digital games, *Entertain. Comput.* 1 (29) (2019) 56–68, <https://doi.org/10.1016/j.entcom.2018.12.001>.
- [23] O.E. Ozenc, User experience and motivation of professional game players: a case study of esports in Turkey, *Game User Experience and Player-Centered Design* (2020) 77–120, [https://doi.org/10.1007/978-3-030-37643-7\\_5](https://doi.org/10.1007/978-3-030-37643-7_5).
- [24] C. Ferguson, R. Lewis, C. Wilks, and R. Picard, The Guardians: Designing a game for long-term engagement with mental health therapy, In *2021 IEEE Conference on Games (CoG)* (2021) (pp. 1–8), IEEE, DOI: 10.1109/CoG52621.2021.9619026.
- [25] M. Copeman, J. Freeman, in: July. Understanding Individual Differences in Mental Health and Games, *Springer Nature Switzerland, Cham*, 2023, pp. 162–175, [https://doi.org/10.1007/978-3-031-35979-8\\_13](https://doi.org/10.1007/978-3-031-35979-8_13).
- [26] S. Paladino, From the 'SelvaOscura' to Paradise Reimagining the Pilgrim's Journey through the Transmedial Realm of Role-Playing Video Games (2021).
- [27] M. Barr, The Force Is Strong with This One (but Not That One): What Makes a Successful Star Wars Video Game Adaptation? In *Arts Vol. 9 No. 4* (2020), p. 131, MDPI, DOI: 10.3390/arts9040131.
- [28] D. Gonçalves, A. Rodrigues, M.L. Richardson, A.A. de Sousa, M.J. Proulx, T. Guerreiro, Exploring asymmetric roles in mixed-ability gaming, in: In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, 2021, pp. 1–14, <https://doi.org/10.1145/3411764.3445494>.
- [29] A. Aldea, Video Games as a Catalyst for Empathy and Perspective Taking, *Plan II Honors Theses-Openly Available* (2019) <https://doi.org/10.26153/tsw/2519>.
- [30] S. Adžić, J. Al-Mansour, H. Naqvi, S. Stambolić, The impact of games on students' educational outcomes, *Entertain. Comput.* 1 (38) (2021) 100412, <https://doi.org/10.1016/j.entcom.2021.100412>.